

MD RAYYAN

B.Tech CSE (Data Science) · NMIMS University, Hyderabad · 2023–2027

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EDUCATION

STME, NMIMS University, Hyderabad

2023 – 2027

B. Tech, Computer Science and Engineering (Data Science)

CGPA: 3.51 / 4.0

CORE TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, HTML/CSS
- **ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV; CNNs, Transfer Learning, Grad-CAM
- **LLM / GenAI:** LangChain, RAG, ChromaDB, Prompt Engineering, Multi-LLM Orchestration
- **Data & Analysis:** Pandas, NumPy, Power BI
- **Frameworks:** React, Next.js, Flask, FastAPI, Streamlit
- **Databases:** MySQL, Firebase, Supabase
- **Cloud & DevOps:** AWS, GCP, Docker, Git, GitHub, MLflow

PROJECTS

DermAegis AI — Skin Lesion Intelligence Platform

[GitHub](#) | [Demo](#) | 2026

Python · TensorFlow · FastAPI · React · Docker · Grad-CAM

- Engineered a 7-class dermoscopic lesion classifier on HAM10000 (10k+ images) using EfficientNetB3 transfer learning, achieving 85.75% accuracy and 0.72 macro F1 on a lesion-grouped split (no image leakage), via two-phase fine-tuning, mixup, label smoothing, and TTA ×4.
- Resolved severe class imbalance and near-doubled melanoma detection (F1 0.33 → 0.61) through upsample-median oversampling and calibrated confidence scoring, benchmarking against MobileNetV2 and ResNet50.
- Built a FastAPI microservice with Monte Carlo Dropout uncertainty, Grad-CAM heatmaps, and an automated Fitzpatrick skin-tone bias detector; deployed via React UI, Streamlit, and Docker Compose.

Argus — Spot-Resilient ML Training Orchestrator (2-person team)

[GitHub](#) | 2026

Python · PyTorch · FastAPI · Docker · MLflow · AWS (S3/ECR)

- Built the ML prediction layer of a system forecasting AWS EC2 Spot instance interruptions ahead of AWS's 2-minute reclamation notice, enabling proactive checkpointing for long-running training jobs.
- Engineered a feature pipeline from raw EC2 Spot price history (rolling means/std, price deltas, temporal encodings) and trained a Transformer encoder with Focal Loss to handle severe interruption-class imbalance, tracked via MLflow.
- Built a Dockerized, load-tested FastAPI /predict service (risk scores per instance-type/AZ) pushed to AWS ECR, integrating with a teammate's Kubernetes Operator via a strict API/data contract for independent parallel development.

BharatHealth Analyst — LLM Agent for Indian Public Health Data Analysis

[GitHub](#) | 2026

Python · LangChain · FastAPI · ChromaDB · Multi-LLM (Grok/Gemini/Groq/OpenRouter)

- Built a 6-tool LangChain ReAct agent enabling natural-language querying over India's NFHS-5 health survey (706 districts, 107 indicators, 36 states/UTs), with semantic search, correlation analysis, and Plotly visualizations.
- Designed BharatHealth-Bench, a first-of-its-kind 200-question benchmark for Indian health-data LLM agents across 5 metrics, including two novel ones (Answer Faithfulness, Reasoning Chain Quality), achieving 90% execution accuracy and 0% hallucination rate.
- Engineered a production-grade architecture with RestrictedPython sandboxing, ChromaDB vector search, and a provider-agnostic multi-LLM fallback layer, exposed via a 10-endpoint FastAPI backend serving a real-time frontend.

ACHIEVEMENTS AND POSITIONS OF RESPONSIBILITY

- **Winner — NMIMS Hackathon;** led development of a scalable digital growth platform for small businesses among 30 competing teams. (2025)
- **Head — Code IT Club;** leading technical strategy and organizing 10+ coding events including webathons and hackathons, reaching 500+ students. (2025–Present)
- **Student Engagement Lead — ELGE Club;** organized 3+ industry-focused workshops, driving 400+ Google Cloud certification sign-ups among students. (2025)
- **Student Ambassador — Google Cloud Foundations;** represented the program across 7 campuses, engaging 500+ students on cloud certifications. (2024–2025)
- **Plantora — AI-Based Smart Agriculture System;** ranked among top 20% at NMIMS Research Symposium. (2026)
- **Finalist — Joy of Programming;** placed among top 6 finalists out of 40+ participants in competitive programming. (2024)

CERTIFICATIONS

- AWS Academy Graduate — ML Foundations & Cloud Foundations
- Google Cloud Computing Foundations + Skill Badges (Network Security, ML Data Prep, Load Balancing)
- Cisco Networking & OS Basics | NPTEL (Python, DBMS)

Credentials: [credly.com/users/rayyan-md](https://www.credly.com/users/rayyan-md)